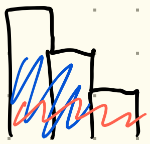
84 - Largest Rectangle In Histogram

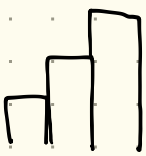


(height, left_starting)

(3,0) initial height

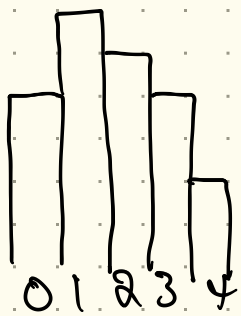
~~(3,0)~~ (2,0) pop (3,0), compute largest rectangle

$3 \cdot (1-0)$



(1,0), (2,1), (3,2)

0 1 2 3 4
4, 6, 5, 4, 2



(4,0)

(4,0), ~~(6,1)~~

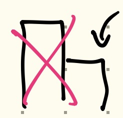
(4,0), (5,1)

(4,0) (4,0) better than (4,3)

(2,4)

Calculate the area of every candidate rectangle

- Monotonically decreasing stack [1, 2, 3]



left = 0

Each rectangle consists of

- left : either the index or the leftmost left it got from popping
- right : determined when a shorter rectangle is encountered
- height : height[i]

Append 0 to height for easy calc of